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^{68}Ga-PSMA-11 Positron Emission Tomography Detects Residual Prostate Cancer after Prostatectomy in a Multicenter Retrospective Study

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Abbreviations and Acronyms

ADT = androgen deprivation therapy

CT = computerized tomography

ISUP = International Society of Urological Pathology

MRI = magnetic resonance imaging

PCa = prostate cancer

PET = positron emission tomography

PROMISE = Prostate Cancer Molecular Imaging Standardized Evaluation

PSMA = prostate specific membrane antigen

PSMA-ligand PET = ^{68}Ga-PSMA-11 PET

RP = radical prostatectomy

Tr = prostate bed

Purpose: Prostate specific antigen persistence after radical prostatectomy is associated with adverse outcomes in patients with prostate cancer. We sought to define regions at risk for residual disease as well as the accuracy of prostate specific membrane antigen ligand positron emission tomography in patients with prostate specific antigen persistence.

Materials and Methods: At 6 participating centers a total of 191 patients who underwent ^{68}Ga-prostate specific membrane antigen-11 positron emission tomography/computerized tomography or positron emission tomography/magnetic resonance imaging for persistently elevated postoperative prostate specific antigen (0.1 ng/ml or greater) were retrospectively included in study. The detection rate and the positive predictive value were determined. In 33 patients with additional prostate specific membrane antigen ligand positron emission tomography before prostatectomy we also determined the rate of positron emission tomography based persistence and recurrence.

Results: Prostate specific membrane antigen ligand positron emission tomography localized prostate cancer in 130 of 191 patients (68%) with prostate specific

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antigen persistence at a median prostate specific antigen of 1.1 ng/ml. The detection rate significantly increased with prostate specific antigen ($p < 0.001$). Regarding prostate specific membrane antigen positron emission tomography/computerized tomography only 61 of 173 patients (35%) had disease confined to the pelvis while 57 of 173 (33%) had distant lesions. The most frequently affected nodal regions were the obturator in 42% and the presacral/mesorectal region in 40%. In 15 of the 33 patients (45%) with prostate specific membrane antigen ligand positron emission tomography before and after surgery at least 1 lesion was detected at baseline (positron emission tomography persistence), 8 (24%) had new lesions (positron emission tomography recurrence) and 10 (30%) had negative positron emission tomography findings. The positive predictive value of prostate specific membrane antigen ligand positron emission tomography was 91%. Systemic therapy initiation was significantly associated with distant lesions on prostate specific membrane antigen ligand positron emission tomography.

Conclusions: Prostate specific membrane antigen ligand positron emission tomography localized prostate cancer in more than two-thirds of patients with high risk features and prostate specific antigen persistence after prostatectomy. Obturator and presacral/mesorectal nodes are at high risk for persistent metastasis.

Key Words: prostatic neoplasms, prostate specific antigen, diagnostic imaging, neoplasm metastasis, positron-emission tomography

MEN with persistently elevated PSA 0.1 ng/ml or greater more than 6 weeks after RP show less favorable cancer control and survival rates with time.[1,2] Precise mapping of initial and residual disease is crucial to determine the extent of local therapy and guide subsequent management as early as possible. The diagnostic value of conventional imaging in this setting is limited. MRI, CT, bone scan, fluorodeoxyglucose or NaF PET/CT are inadequate below optimal therapeutic PSA thresholds during biochemical failure after radical surgery.[3]

PSMA-ligand PET images PSMA expression and detects PCa with unprecedented accuracy before and after prostatectomy.[4–7] In a recent prospective multicenter trial designed for regulatory approval PSMA-ligand PET demonstrated a high detection rate and positive predictive value for localizing recurrent PCa.[8] However, to our knowledge the capability of PSMA-ligand PET to localize lesions before and after RP has not been assessed systematically in patients with PSA persistence. Lesion localization would identify regions at high risk for persistent metastasis with the ultimate aim of more effective initial and subsequent therapy.

We hypothesized that PSMA-ligand PET would identify disease with high accuracy and PET based mapping of disease would have implications for initial and subsequent management. Therefore, the aim of this study was to identify regions at risk for residual disease in patients with PSA persistence after prostatectomy. We further aimed to define PSMA PET accuracy and interreader agreement in this setting.

MATERIALS AND METHODS

Patient Population

Between August 2011 and May 2018 a total of 191 patients a median of 67 years old (IQR 61–71) with PCa and PSA persistence after RP at a total of 6 high volume nuclear medicine centers were retrospectively included in study (supplementary material, https://www.jurology.com). Study inclusion criteria were 1) RP of PCa, 2) persistently elevated postoperative PSA (0.1 ng/ml or greater) 6 weeks or more after surgery and 3) PSMA-ligand PET performed within 12 months after RP. Subgroup analysis was performed in 33 patients with PSMA-ligand PET before RP for preoperative staging and PSMA-ligand PET after RP for PSA persistence. Lesion validation was recorded as well as post-PET management when available. Anonymized data were centrally collected at the Department of Nuclear Medicine, University Duisburg-Essen, Essen, Germany. This retrospective analysis was approved by the local ethics committee (IRB No. 18-8136-BO), which waived requirement to obtain informed consent for inclusion.

Images

Acquisition. ^{68}Ga-PSMA-11 (Glu-NH-CO-NH-Lys-(Ahx)-[^{68}Ga(HBED-CC)]) was used as the PSMA ligand. Median injected activity was 156 MBq (IQR 128–189). The median uptake period was 63 minutes (IQR 57–74). Images were acquired by PET/CT in 173 patients (91%) and PET/MRI in 18 (9%). All PET/MRI and 91 of the 173 PET/CT scans (53%) were acquired after intravenous contrast administration. PET protocols were in accordance with the Joint SNMMI (Society of Nuclear Medicine and Molecular Imaging)/EANM (European Association on Nuclear Medicine) procedure guideline for PSMA-ligand PET imaging.[9]

Interpretation. Anonymized imaging data sets were evaluated independently by 3 experienced nuclear medicine physicians who followed recent recommendations[9,10] and were blinded to clinical history. Readers were trained with a PSMA-ligand PET course of 30 cases including routine findings (ranging from unremarkable to extensive disease) and typical PSMA-ligand PET pitfalls before analyzing images.[11] Based on the PROMISE molecular imaging TNM system the systematically analyzed regions (subregions) were the prostate bed,

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pelvic nodes (N) (internal iliac, obturator, external iliac, mesorectal, presacral and common iliac), extrapelvic nodes (M1a) (retroperitoneal, mesenteric, inguinal and above diaphragm), bone (M1b) and visceral organs (M1c).[12,13]

Lesion Validation

Local investigators reviewed patient files for correlative and followup information acquired during routine clinical practice. Lesions were confirmed by histopathology and/or biopsy, lesion presence on correlative CT, MRI and/or bone scan, or a change in size, disappearance and/or appearance on followup CT, MRI and/or bone scan. The local investigators interpreted the composite reference standard after reviewing followup information. Positive PET findings were validated as positive or false-positive on a regional basis.

Statistics

We assessed PROMISE TNM separately for PET/CT and PET/MRI. All other analyses were done using pooled PET/CT and PET/MRI findings. The PSMA-ligand PET detection rate was calculated on a patient basis stratified by PSA at the time of imaging and the PSA nadir. PSA nadir classes were determined in accordance with the study by Wiegel et al.[14]

Descriptive statistics were calculated. We used the nonparametric Mann-Whitney test, the Student t-test and the chi-square test as appropriate to compare the distribution of patient clinical characteristics across PET results. To investigate the association between patient characteristics and the presence of lesions we performed univariable and multivariable logistic regression analyses with inclusion in multivariable regression at $p < 0.05$. PSA at the time of the scan was coded as a continuous variable, and patient age (less than 65 vs 65 years or greater), ISUP grade (less than 4 vs 4 or greater), tumor stage (less than T3 vs T3 or greater) and ongoing ADT (yes vs no) were coded as categorical variables. To test goodness of fit we applied the Hosmer-Lemeshow test. For binary data agreement among the observer groups was evaluated by the Fleiss κ. Statistical analyses were done with R, version 3.5.1 (https://www.r-project.org/foundation/) for agreement and with STATA® 15 for all other analyses.

RESULTS

Patient Characteristics

Table 1 lists patient characteristics. All patients had pathological risk features after RP, which were ISUP Grade Group 4 or greater in 58%, pathological stage T3a or greater in 77%, N1 in 43% and/or R1 in 42%. The median PSA nadir after RP was 0.7 ng/ml (IQR 0.3–2.4) and median PSA at PSMA-ligand PET was 1.1 ng/ml (IQR 0.5–3.7). No patient underwent radiotherapy before PET.

Detection Rate and Lesion Location

PSMA-ligand PET localized disease in 130 of 191 patients, resulting in a 68% overall detection rate. The probability of detecting any lesion or multiple lesion sites was associated with the PSA level and with PSA at the time of PET ($p < 0.001$, fig. 1). On patient based analysis of PSMA-ligand PET/CT in 173 patients disease confined to the pelvis was detected in 61 of 173 patients (35%) with predominant N1 disease in 48 (28%) and local recurrence in only 9 (5%). Overall the most frequently affected pelvic nodal regions were the obturator region in 47% of cases and the presacral/mesorectal region in 40% (table 2).

Distant disease (extrapelvic lymph node, bone and/or viscera) was detected in 57 of the 173 patients (33%). The most common M1 location was bone, noted in 27 of 173 cases (16%), and extrapelvic lymph nodes, noted in 16 (9%). PSMA-ligand PET/MRI detected disease in 18 cases (67%). The supplementary material (https://www.jurology.com) presents the details. Most PET positive nodes presented as intense tracer uptake. However, they were not enlarged according to CT/MRI criteria (supplementary material, https://www.jurology.com). PSMA-ligand PET detected disease confined to the pelvis in 26 and distant disease in 19 of the 71 patients (37% and 27%, respectively) with N0 disease postoperatively (supplementary material, https://www.jurology.com).

To assess a potential difference in the 12-week vs the 12-month times we performed subgroup analysis including only the 36 patients in whom PSMA-ligand PET was done within 12 weeks postoperatively. We found a 69% detection rate, which was in line with the overall 68% detection rate in the entire cohort (supplementary material, https://www.jurology.com). On subgroup analysis median PSA at the time of PET was 1.7 ng/ml, slightly higher than the 1.1 ng/ml in the entire cohort. This resulted in a higher proportion of patients with M1 (systemic) disease, ie 16 of 36 or 44% vs 33% of the entire cohort.

Table 1. *Characteristics of 191 patients*

Mean ± SD age/median (IQR)	66 ± 8/67	(61–71)
Mean ± SD ng/ml PSA/median (IQR):		
Initial	26.9 ± 37.0/12	(8.3–29)
Nadir	2.7 ± 5.7/0.7	(0.3–2.4)
At PET	4.5 ± 12.8/1.1	(0.5–3.7)
Mean ± SD mos change PET − RP/median (IQR)	5.0 ± 2.8/4	(3–7)
No. pts/total No. (%):		
T3a or greater	130/170	(77)
N1	54/125	(43)
R1	63/120	(42)
ISUP Grade Group 4 or greater	105/180	(58)
No. ADT/total No. (%):		
Previous	35/191	(18)
At scan	24/189	(13)
No. PET/total No. (%)		
MRI	18/191	(9)
CT	173/191	(91)
No./total No. (%) RP technique:		
PLND	182/191	(95)
No PLND	9/191	(5)

Histopathology data were not available on all patients.

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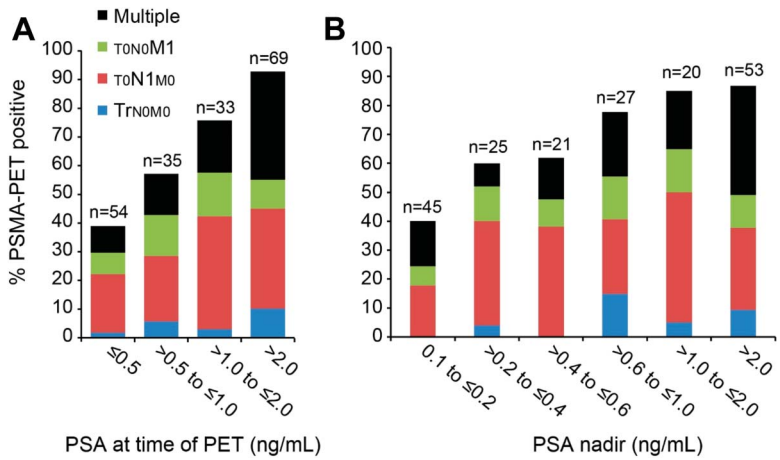

Figure 1. PSMA-ligand PET detection rate on patient basis stratified by PSA at PET (*A*) and at PSA nadir (*B*). PSA nadir ranges were determined in accordance with Wiegel et al.[14] Portion of patients with PSMA PET positive findings were grouped by disease location in accordance with PROMISE. *N1*, pelvic nodes. *M1*, extrapelvic. *Multiple*, multiple regions (black bars). Green bars indicate T0N0M1. Pink bars indicate T0N1M0. Blue bars indicate TrN0M0.

Positron Emission Tomography Based Lesion Persistence vs Recurrence in Patient Subset with PSMA PET before and after Radical Prostatectomy

In a subgroup of 33 patients PSMA-ligand PET was performed within 2 months before and a median of 4 months after RP. All lesions noted on PET before RP were reported for RP planning. Two patients had focal bone uptake, which was rated equivocal on the local PET read, and surgery was planned. However, bone uptake was rated as positive for metastatic disease on blinded consensus reads off site during our study analysis.

In 15 of the 33 patients (45%) at least 1 lesion was identified before RP, which was still detected after RP. Of the 33 patients 8 (24%) had only persistent lesion(s) while 7 (21%) had mixed lesion(s), defined as at least 1 persistent lesion and at least 1 new lesion after RP (table 3). The most frequently affected nodal regions with PET persistence were the obturator region (2 of 12) and the presacral/mesorectal region (7 of 12 N1 regions). Lesions recurred most often in the internal iliac region and the common iliac region (6 and 5 of 17 cases, respectively) (fig. 2 and supplementary material, https://www.jurology.com). The supplementary material, https://www.jurology.com) also provides patient examples of persistent lymph node metastasis.

Table 2. *Stage by PSMA-ligand PET/CT PROMISE in 173 patients*

Molecular Imaging TNM Stage	No. PET/CT Read Consensus (%)
T0N0M0 (no PET lesions)	55 (32)
Tr/N1 (local):	61 (35)
TrN0M0	9 (5)
T0N1M0	48 (28)
TrN1M0	4 (2)
M1 (systemic) :	57 (33)
T0N0M1	25 (14)
TrN0M1	1 (1)
T0N1M1	27 (16)
TrN1M1	4 (2)
N1 disease subregions:	83 (48)
Common iliac	31 (37)
External iliac	26 (31)
Internal iliac	25 (30)
Obturator	39 (47)
Presacral/mesorectal	33 (40)
Other	1 (1)
M1 disease subregions:	57 (33)
M1a (lymph node only)	16 (28)
M1b (bone only)	27 (47)
M1c (visceral only)	1 (2)
M1 (multiple)	13 (23)
TNM disease extent (No. sites):	118 (68)
Unifocal (1)	45 (38)
Oligometastatic (2-3)	28 (24)
Multiple/disseminated (4 or more)	45 (38)

Interobserver Agreement and Lesion Validation

Overall interpretation among 3 readers showed substantial to almost perfect agreement (Fleiss $\kappa = 0.79$). Agreement was substantial for the local prostate bed ($\kappa = 0.67$) and almost perfect for the pelvic nodes ($\kappa = 0.89$) and distant metastasis ($\kappa = 0.81$ for M1ac disease and 0.92 for M1b disease).

Table 3. *PSMA-ligand PET findings in PET preRP and post-RP subgroups in 33 patients*

	No. PET (%)			
PCa Location	Pos Post-RP	Persistence	Recurrence	Mixed
Overall	23 (70)	8 (24)	8 (24)	7 (21)
TrN0M0	2 (6)	2 (6)	0	0
T0N1M0	12 (36)	4 (12)	6 (18)	2 (6)
T0N0M1a	1 (3)	0	1 (3)	0
T0N0M1b	3 (9)	2 (6)	1 (3)	0
T0N0M1c	0	0	0	0
Multiple	5 (15)	0	0	5 (15)

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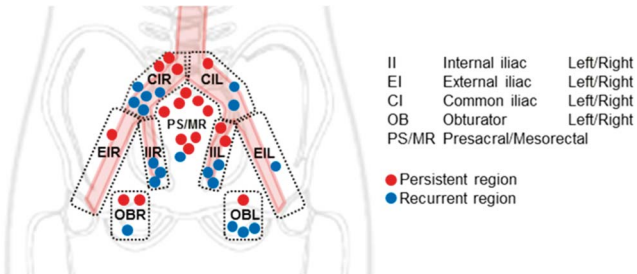

Figure 2. PSMA-ligand PET findings in PET pre-RP and post-RP subgroup of 33 patients. *II,* left/right internal iliac. *EI,* left/right external iliac. *CI,* left/right common iliac. *OB,* left/right obturator. *PS/MR,* presacral mesorectal. Red dots represent disease persistence on PET. Blue dots represent disease recurrence on PET.

Validation was available in a total of 53 regions in 43 patients. Of the regions 48 were positive, including 8 with biopsy/histology and 40 with imaging followup, resulting in a 91% positive predictive value. Five regions were false-positive, including 2 pelvic regions negative on histopathology/biopsy and 3 pelvic regions negative on correlative MRI.

Factors Associated with Local vs Distant Disease

On multivariable regression analysis PSA at the time of PET was associated with the presence of Tr/N1 (OR 1.84, CI 95% 1.28–2.65, p=0.001) and M1 disease (OR 1.74, 95% CI 1.22–2.50, p=0.002). Similarly ISUP grade 4 or greater disease was associated with the presence of Tr/N1 disease (OR 2.45, 95% CI 1.09–5.49, p=0.03) or M1 disease (OR 3.11, 95% CI 1.12–8.65, p=0.03). Stage T3a or greater was associated with the presence of M1 disease (OR 7.64, 95% CI 1.66–35.08, p=0.01). All other factors were not significantly associated with a disease pattern (table 4).

DISCUSSION

After RP and pelvic lymph node dissection serum PSA is expected to reach an undetectable level of less than 0.1 ng/ml within 21 to 30 days based on the half-life of about 3 days.[15] Persistently detectable PSA after RP has been attributed to the presence of micrometastatic or metastatic disease, or residual prostate tissue. More important is that in several studies postoperative PSA persistence was associated with clinical recurrence and short survival.[2,14,16–19] Thus, postoperative local and/or systemic treatments are offered and tailored according to the individual risk.[2] In a large study of persistently elevated PSA after RP Gandaglia et al found an association of postoperative radiation therapy with a survival benefit in men with pathological high risk features.[20] Accurate disease localization is critical for effective local treatment.

In our study PSMA-ligand PET localized PCa in more than two-thirds of the patients with PSA persistence. These findings are in line with those of Ceci et al, who previously reported a 65% detection rate in a smaller patient subgroup.[21] In our study disease localization was characterized by 91% positive predictive value and high interobserver agreement. Stage was associated with subsequent treatment pathways. We identified regions at risk for residual disease, including obturator nodes, presacral/mesorectal nodes and extrapelvic lesions commonly outside the resection template. More importantly, in a subcohort we found that lesions had already been noted on PET in about half of the patients before surgery. Thus, in patients with anticipated high risk features preoperative PET has the potential to improve the selection of candidates for more extended or PSMA radioguided surgery.[22] The prospective, open label, randomized, phase III proPSMA trial will provide data to establish whether PSMA-PET/CT should replace conventional imaging in the primary staging of select patients with high risk, localized PCa.[23]

Our findings come with implications for the treatment of persistent disease. At biochemical failure early salvage radiotherapy is recommended, which potentially provides a survival benefit or even cure.[20,24] More than 60% of patients treated before

Table 4. *Logistic regression analysis of Tr/N1 or M1 disease in 191 patients*

	Simple Logistic		Multivariable Logistic	
Disease	OR (95% CI)	p Value	OR (95% CI)	p Value
Tr/N1:				
PSA at PET	1.98 (1.35–2.90)	0.001	1.84 (1.28–2.65)	0.001
Age 65 or greater	1.28 (0.64–2.59)	0.48	—	—
ISUP Grade Group 4 or greater	2.09 (1.01–4.32)	0.047	2.45 (1.09–5.49)	0.03
T3a or greater	1.64 (0.73–3.66)	0.23	—	—
ADT at scan	0.58 (0.17–1.94)	0.38	—	—
M1:				
PSA at PET	1.94 (1.34–2.80)	0.001	1.74 (1.22–2.50)	0.002
Age 65 or greater	0.88 (0.43–1.81)	0.73	—	—
ISUP Grade Group 4 or greater	4.71 (2.01–10.57)	0.001	3.11 (1.12–8.65)	0.03
T3a or greater	5.92 (2.00–17.56)	0.001	7.64 (1.66–35.08)	0.01
ADT at scan	1.93 (0.70–5.31)	0.20	—	—

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PSA increased to greater than 0.5 ng/ml will achieve undetectable PSA and show more favorable outcomes, as reported by Stephenson[25] and Tendulkar[26] et al. In our study about 30% of the patients had PSA 0.5 ng/ml or less at PET and, thus, they were good candidates for salvage radiotherapy.[25] The overall PSMA-ligand PET detection rate was around 40% with only pelvic nodal disease in about half of the patients (fig. 1).

Also, PSMA-ligand PET reveals targetable local or distant disease each in about a third of patients with initial stage pN0 disease. PET information considerably impacts the radiotherapy plan, as noted by Calais et al in a retrospective study in 270 patients.[27] The potential clinical benefit of PSMA-ligand PET guidance is under investigation in a phase 3, randomized trial aimed to improve the oncologic outcome currently being performed at UCLA.[27] In a retrospective study of the impact of PSMA-ligand PET/CT on the radiotherapeutic approach to PCa Schmidt-Hegemann et al found PSA persistence in 68 of 88 men (77% detection rate) with a median PET PSA of 1.2 ng/ml.[28] As in our cohort, most patients had lesions outside the prostate bed, leading to post-PET expansion of radiotherapy in all patients with positive PET findings. To expand this concept distant lesions can also be targeted for ablation with the intent to postpone ADT.[29,30]

Our findings come with implications for clinical trial design. Regulatory agencies and scientific reviewers demand implementation of the histopathology reference standard. We found that surgery/histopathology is blind to disease outside the resection template. PSMA-ligand PET demonstrates a high detection rate, high reproducibility and a high positive predictive value[8] for whole body prostate cancer localization. Thus, PET should be included more stringently in study protocols to guide and complement tissue sampling for improved disease validation.

Limitations of our study include the retrospective design, the more than 12-week period between RP and PSMA-ligand PET, and the low validation rate for PET positive lesions. Due to a delay between the biochemical persistence diagnosis and PET the reported PSA and PET detection rate was likely higher than expected at the time of the biochemical persistence diagnosis.

CONCLUSIONS

We retrospectively analyzed a population with PCa who had pathological high risk features and PSA persistence after RP. PSMA-ligand PET localized disease in more than two-thirds of the patients. Obturator nodes, presacral/mesorectal nodes and extrapelvic regions frequently harbored persistent prostate cancer. PSMA-ligand PET should be offered to localize persistent disease in patients at high risk.

REFERENCES

1. Rogers CG, Khan MA, Craig Miller M et al: Natural history of disease progression in patients who fail to achieve an undetectable prostate-specific antigen level after undergoing radical prostatectomy. Cancer 2004; **101:** 2549.
2. Bianchi L, Nini A, Bianchi M et al: The role of prostate-specific antigen persistence after radical prostatectomy for the prediction of clinical progression and cancer-specific mortality in node-positive prostate cancer patients. Eur Urol 2016; **69:** 1142.
3. Vargas HA, Martin-Malburet AG, Takeda T et al: Localizing sites of disease in patients with rising serum prostate-specific antigen up to 1ng/ml following prostatectomy: how much information can conventional imaging provide? Urol Oncol 2016; **34:** 482.e5.
4. Herlemann A, Wenter V, Kretschmer A et al: 68Ga-PSMA positron emission tomography/computed tomography provides accurate staging of lymph node regions prior to lymph node dissection in patients with prostate cancer. Eur Urol 2016; **70:** 553.
5. Eiber M, Maurer T, Souvatzoglou M et al: Evaluation of hybrid ^{68}Ga-PSMA ligand PET/CT in 248 patients with biochemical recurrence after radical prostatectomy. J Nucl Med 2015; **56:** 668.
6. Perera M, Papa N, Roberts M et al: Gallium-68 prostate-specific membrane antigen positron emission tomography in advanced prostate cancer-updated diagnostic utility, sensitivity, specificity, and distribution of prostate-specific membrane antigen-avid lesions: a systematic review and meta-analysis. Eur Urol 2019; doi: 10.1016/j.eururo.2019.01.049.
7. Afshar-Oromieh A, Avtzi E, Giesel FL et al: The diagnostic value of PET/CT imaging with the (68) Ga-labelled PSMA ligand HBED-CC in the diagnosis of recurrent prostate cancer. Eur J Nucl Med Mol Imaging 2015; **42:** 197.
8. Fendler WP, Calais J, Eiber M et al: Assessment of 68Ga-PSMA-11 PET accuracy in localizing recurrent prostate cancer: a prospective single-arm clinical trial. JAMA Oncol 2019; **5:** 856.
9. Fendler WP, Eiber M, Beheshti M et al: 68Ga-PSMA PET/CT: Joint EANM and SNMMI procedure guideline for prostate cancer imaging: version 1.0. Eur J Nucl Med Mol Imaging 2017; **44:** 1014.
10. Rauscher I, Maurer T, Fendler WP et al: (68)Ga-PSMA ligand PET/CT in patients with prostate cancer: how we review and report. Cancer Imaging 2016; **16:** 14.
11. Fendler WP, Calais J, Allen-Auerbach M et al: 68Ga-PSMA-11 PET/CT interobserver agreement for prostate cancer assessments: an international multicenter prospective study. J Nucl Med 2017; **58:** 1617.
12. Eiber M, Herrmann K, Calais J et al: Prostate Cancer Molecular Imaging Standardized Evaluation (PROMISE): proposed miTNM classification for the interpretation of PSMA-ligand PET/CT. J Nucl Med 2018; **59:** 469.
13. Taylor A, Rockall AG and Powell ME: An atlas of the pelvic lymph node regions to aid radiotherapy target volume definition. Clin Oncol (R Coll Radiol) 2007; **19:** 542.
14. Wiegel T, Bartkowiak D, Bottke D et al: Prostate-specific antigen persistence after radical prostatectomy as a predictive factor of clinical relapse-free survival and overall survival: 10-year data of the ARO 96-02 trial. Int J Radiat Oncol Biol Phys 2015; **91:** 288.

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15. Partin AW and Oesterling JE: The clinical usefulness of prostate specific antigen: update 1994. J Urol 1994; **152:** 1358.

16. de Boo L, Pintilie M, Yip P et al: Time from first detectable PSA following radical prostatectomy to biochemical recurrence: a competing risk analysis. Can Urol Assoc J 2015; **9:** E14.

17. Audenet F, Seringe E, Drouin SJ et al: Persistently elevated prostate-specific antigen at six weeks after radical prostatectomy helps in early identification of patients who are likely to recur. World J Urol 2012; **30:** 239.

18. Moreira DM, Presti JC, Aronson WJ et al: Definition and preoperative predictors of persistently elevated prostate-specific antigen after radical prostatectomy: results from the Shared Equal Access Regional Cancer Hospital (SEARCH) database. BJU Int 2010; **105:** 1541.

19. Preisser F, Chun FKH, Pompe RS et al: Persistent prostate-specific antigen after radical prostatectomy and its impact on oncologic outcomes. Eur Urol 2019; **76:** 106.

20. Gandaglia G, Boorjian SA, Parker WP et al: Impact of postoperative radiotherapy in men with persistently elevated prostate-specific antigen after radical prostatectomy for prostate cancer: a long-term survival analysis. Eur Urol 2017; **72:** 910.

21. Ceci F, Castellucci P, Graziani T et al: 68Ga-PSMA-11 PET/CT in recurrent prostate cancer: efficacy in different clinical stages of PSA failure after radical therapy. Eur J Nucl Med Mol Imaging 2019; **46:** 31.

22. Maurer T, Robu S, Schottelius M et al: 99mTechnetium-based prostate-specific membrane antigen-radioguided surgery in recurrent prostate cancer. Eur Urol 2019; **75:** 659.

23. Hofman MS, Murphy DG, Williams SG et al: A prospective randomized multicentre study of the impact of gallium-68 prostate-specific membrane antigen (PSMA) PET/CT imaging for staging high-risk prostate cancer prior to curative-intent surgery or radiotherapy (proPSMA study): clinical trial protocol. BJU Int 2018; **122:** 783.

24. Cornford P, Bellmunt J, Bolla M et al: EAU-ESTRO-SIOG guidelines on prostate cancer. Part II: treatment of relapsing, metastatic, and castration-resistant prostate cancer. Eur Urol 2017; **71:** 630.

25. Stephenson AJ, Scardino PT, Kattan MW et al: Predicting the outcome of salvage radiation therapy for recurrent prostate cancer after radical prostatectomy. J Clin Oncol 2007; **25:** 2035.

26. Tendulkar RD, Agrawal S, Gao T et al: Contemporary update of a multi-institutional predictive nomogram for salvage radiotherapy after radical prostatectomy. J Clin Oncol 2016; **34:** 3648.

27. Calais J, Czernin J, Fendler WP et al: Randomized prospective phase III trial of 68Ga-PSMA-11 PET/CT molecular imaging for prostate cancer salvage radiotherapy planning [PSMA-SRT]. BMC Cancer 2019; **19:** 18.

28. Schmidt-Hegemann NS, Eze C, Li M et al: Impact of 68Ga-PSMA-PET/CT on the radiotherapeutic approach for prostate cancer in comparison to CT: a retrospective analysis. J Nucl Med 2019; **60:** 963.

29. Siva S, Bressel M, Murphy DG et al: Stereotactic ablative body radiotherapy (SABR) for oligometastatic prostate cancer: a prospective clinical trial. Eur Urol 2018; **74:** 455.

30. Ost P, Reynders D, Decaestecker K et al: Surveillance or metastasis-directed therapy for oligometastatic prostate cancer recurrence: a prospective, randomized, multicenter phase II trial. J Clin Oncol 2018; **36:** 446.

EDITORIAL COMMENT

There is now considerable interest in identifying the location of PCa in the biochemical recurrence/persistence setting. Previously it was well recognized that any type of conventional imaging in patients with low PSA following radical RP was futile due to the extremely poor sensitivity of CT, MRI and bone scanning at PSA below 1 ng/ml. Therefore, the traditional paradigm has been to offer postoperative radiation to patients with persistent or recurrent PSA after prostatectomy, especially in the context of lower PSA, eg less than 1 ng/ml. This blind strategy still leads to ongoing biochemical recurrence in many patients due to unrecognized disease outside the target area in the pelvis.

In this study the authors report that PSMA PET/CT can identify prostate cancer in 67% of patients with biochemical recurrence at a median PSA of only 1.1 ng/ml. Even at these low levels 33% of patients have distant disease and, therefore, radiotherapy to the pelvis would likely be disappointing. This retrospective study adds to the body of literature supporting the utility of PSMA PET/CT to identify PCa in this setting (reference 6 in article). Moreover, the only FDA (Food and Drug Administration) approved PET tracer to evaluate biochemical recurrence, fluciclovine, has been shown to perform poorly compared with PSMA PET/CT.[1]

However, despite this impressive data and widespread use of PSMA PET/CT (at least in some countries such as Australia and Germany), it is not clear whether identifying lesions at lower PSA leads to an improvement in patient outcomes. In fact, it may encourage a metastasis directed therapy approach, which has not been shown to improve survival or quality of life. Therefore, at least for now we should accept that PSMA PET/CT is the best imaging modality to identify low volume PCa but what we do with that knowledge is not clear.[2]

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REFERENCES

1. Calais J, Ceci F, Eiber M et al: (18)F-fluciclovine PET-CT and (68)Ga-PSMA-11 PET-CT in patients with early biochemical recurrence after prostatectomy: a prospective, single-centre, single-arm, comparative imaging trial. Lancet Oncol 2019; doi: https://doi.org/10.1016/S1470-2045(19) 30415-2.

2. Vapiwala N, Hofman MS, Murphy DG et al: Strategies for evaluation of novel imaging in prostate cancer: putting the horse back before the cart. J Clin Oncol 2019; **37:** 765.

REPLY BY AUTHORS

PSMA PET demonstrates superior diagnostic accuracy for the localization of biochemically recurrent PCa. Disease detection leads to a major change in management.[1] Whether a PSMA PET induced management change at diagnosis or recurrence will translate into improved oncologic outcomes is currently under investigation in several prospective clinical trials (reference 27 in article)[2] and in the PSMA PET Scans in People Prostate Cancer trial (ClinicalTrials.gov NCT04030338).

REFERENCES

1. Calais J, Fendler WP, Eiber M et al: Impact of 68Ga-PSMA-11 PET/CT on the management of prostate cancer patients with biochemical recurrence. J Nucl Med 2018; **59:** 434.

2. Hofman MS, Murphy DG, Williams SG et al: A prospective randomized multicentre study of the impact of gallium-68 prostate-specific membrane antigen (PSMA) PET/CT imaging for staging high-risk prostate cancer prior to curative-intent surgery or radiotherapy (proPSMA study): clinical trial protocol. BJU Int 2018; **122:** 783.

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